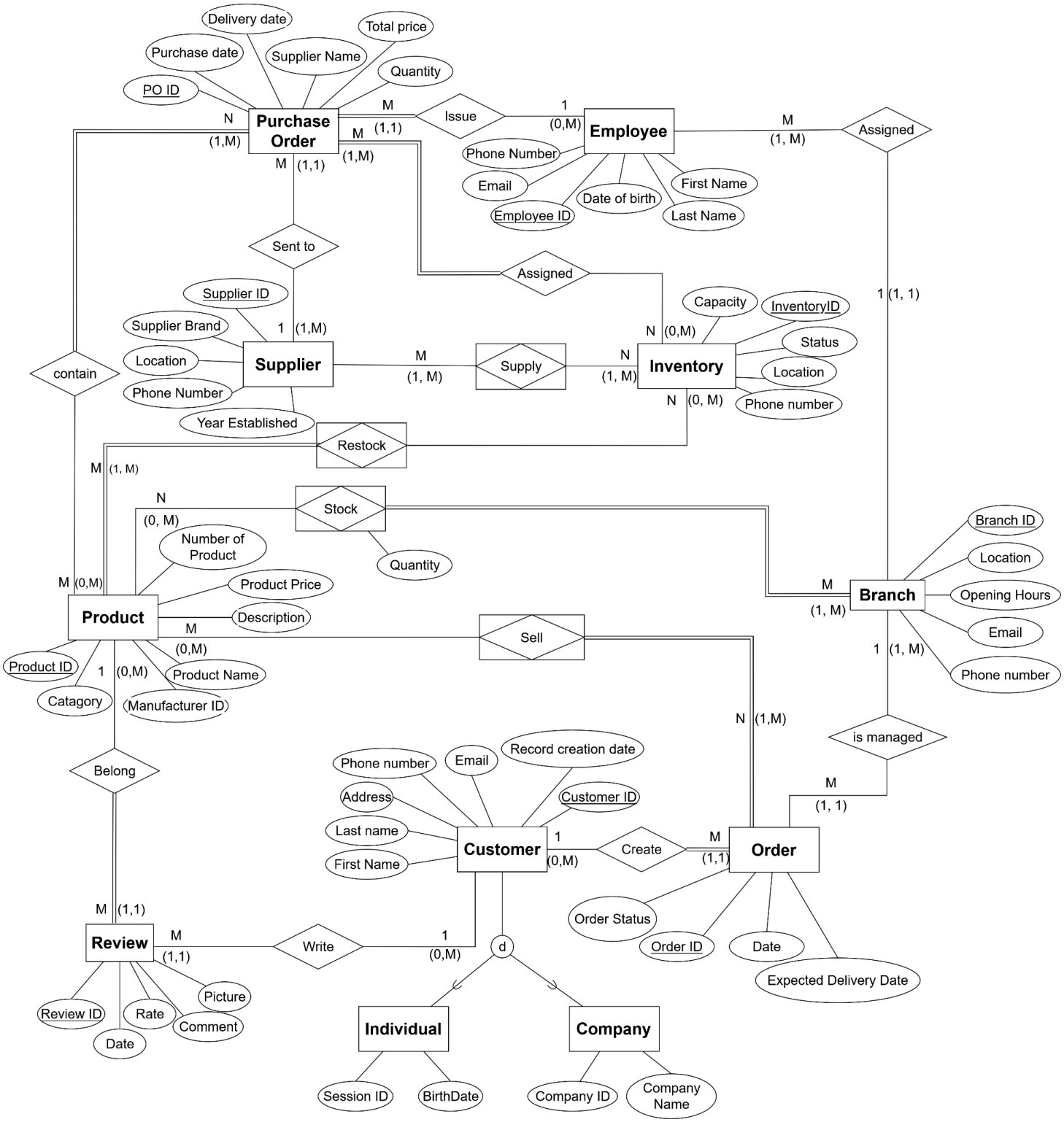
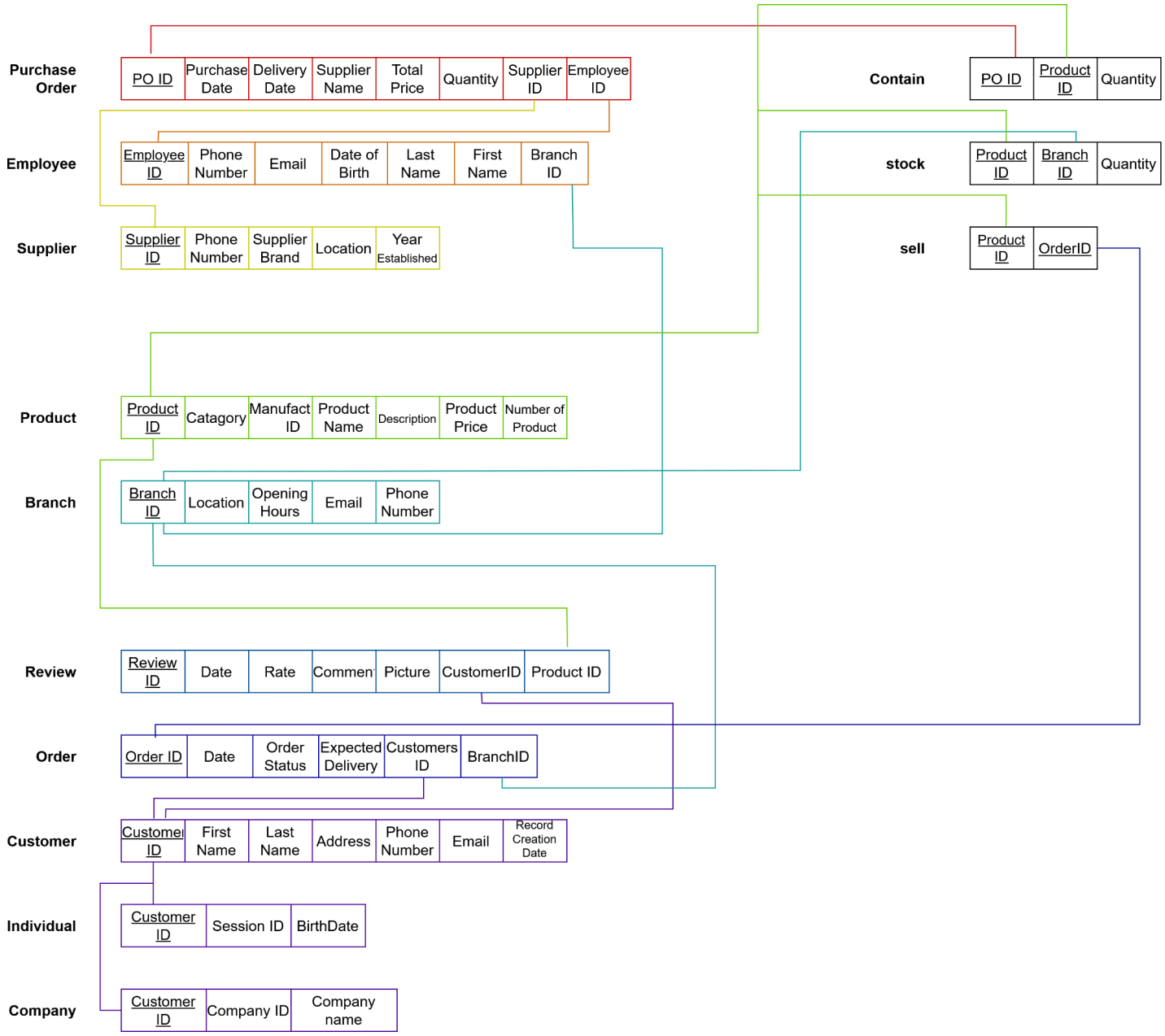
Entity Relationship Diagram

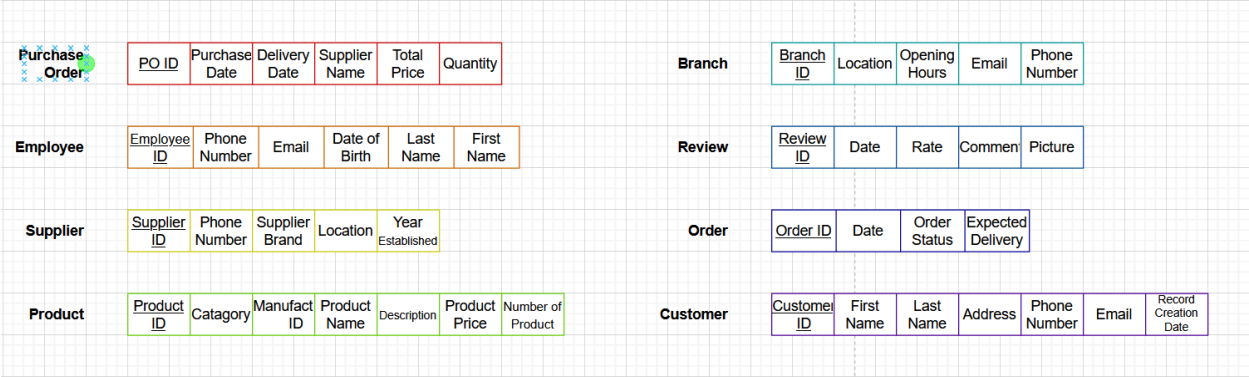


ERD/EERD Transformation



1. Transforming regular entity types

Purchase order, Employee, Supplier, Product, Branch, Review, Order, and Customer.



2. Transforming weak entities types

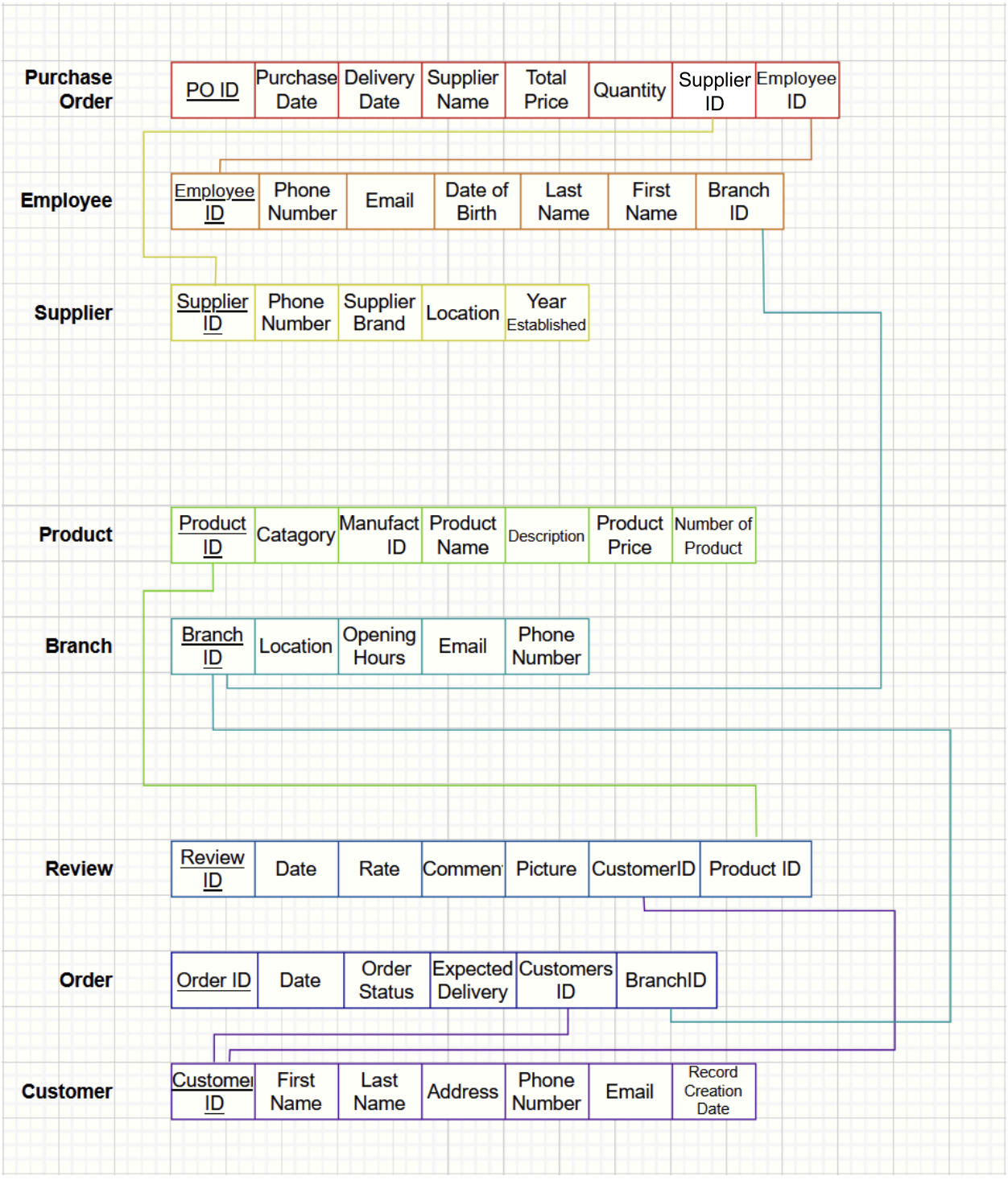
No weak entity.

3. Transforming binary 1:1 relationship types

No 1:1 relationship.

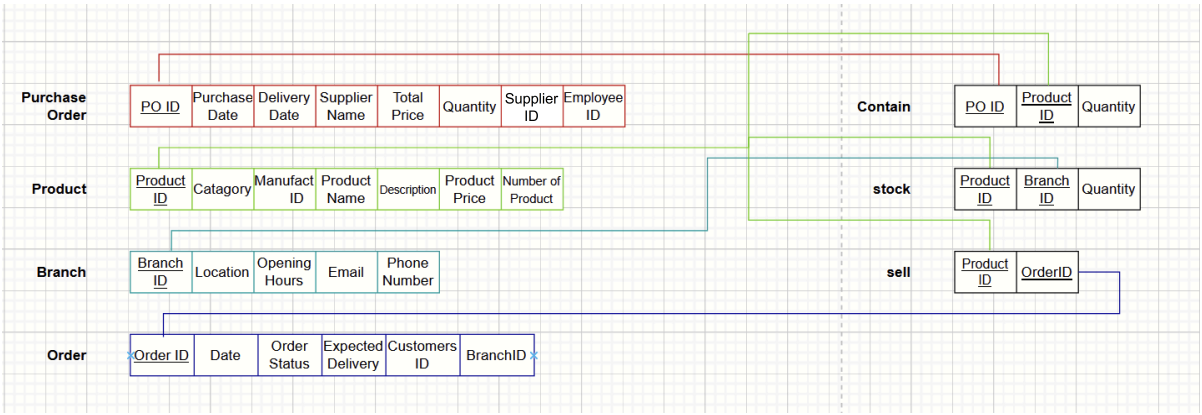
4. Transforming binary 1:N relationship types

Employee issue Purchase order, Purchase order sent to Supplier, Branch assign Employee, Branch manage Order, Customer create Order, Customer write Review



5. Transforming binary M:N relationship types

Purchase order contain Product, Branch sell Product, Order have Product



6. Transforming multi-valued attribute A in relation K

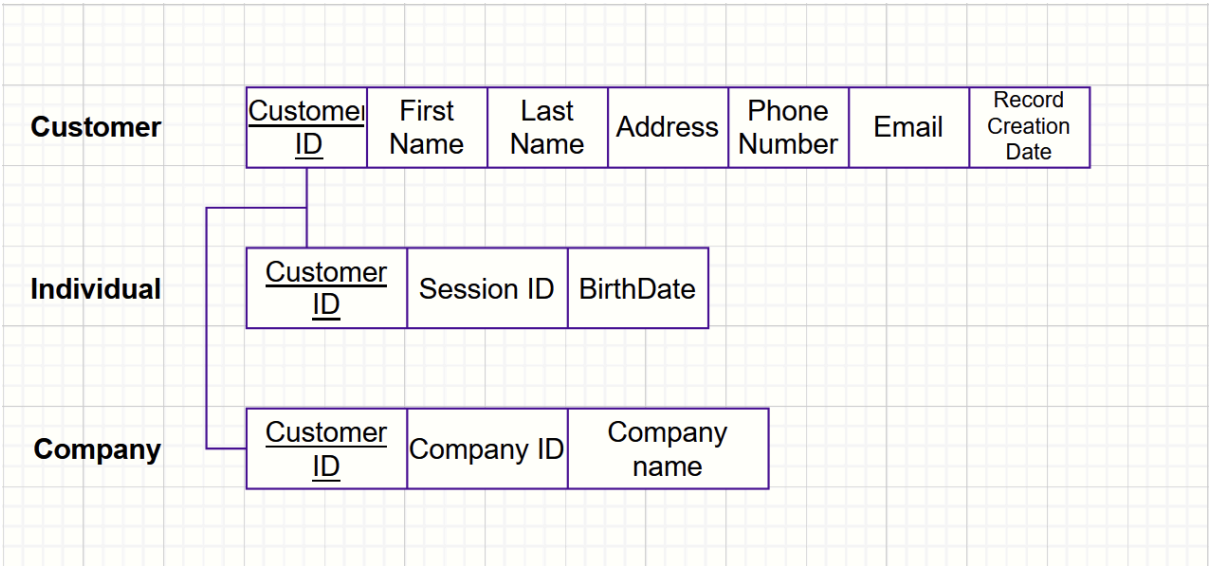
No multi-valued attribute.

7. Transforming n-ary relationship types

No n-ary relationship.

8. Transforming specialization or generalization

Customer contain Member and Guest.

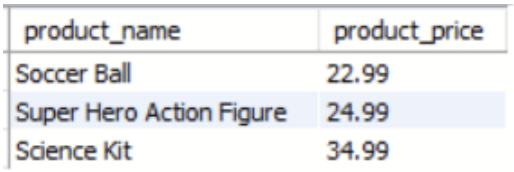


Data Dictionary

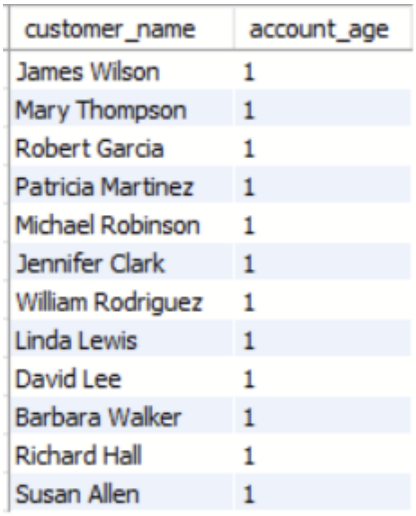
Table Name	Attribute Name	Contents	Type	Format	Nullable	Range	Key	FK Reference Table
purchase_order	po_id	Purchase order ID	varchar(20)	xxxxxxx			PK	
	purchase_date	Purchase Date	datetime	xxxx-xx-xx				
	delivery_date	Delivery Date	datetime	xxxx-xx-xx				
	supplier_name	Supplier Name	varchar(30)	Xxxxxxx				
	total_price	Purchase order Total Price	decimal(9,2)	x,xxx.xx				
	quantity	Purchase Order Quantity	int	xxxx		1 to 9999		
	supplier_id	Supplier ID	varchar(20)	xxxxxxx			FK	Supplier [supplier_id]
	employee_id	Employee ID	varchar(20)	xxxxxxx			FK	Employee [Employee ID]
employee	employee_id	Employee ID	varchar(20)	xxxxxxx			PK	
	phone_number	Employee's Phone Number	varchar(12)	0xx-xxx-xxxx				
	email	Employee's Email	varchar(30)	Xxx@xx.com				
	birth_date	Employee Date of Birth	date	xxxx-xx-xx				
	last_name	Employee's Last name	nvarchar(30)	Xxxxxxx				
	first_name	Employee's First name	nvarchar(30)	Xxxxxxx				
	branch_id	Branch ID	varchar(20)	xxxxxxx			FK	Branch [Branch ID]
supplier	supplier_id	Supplier ID	varchar(20)	xxxxxxx			PK	
	phone_number	Supplier Phone Number	varchar(12)	0xx-xxx-xxxx				
	supplier_name	Supplier Name	varchar(30)	Xxxxxxx				
	location	Supplier's Location	nvarchar(100)	Xxxxxxx				
	year_established	Supplier's established year	date	xxxx-xx-xx				

product	product_id	Product ID	varchar(20)	xxxxxx			PK	
	category	Product's category	varchar(20)	Xxxxxxx				
	manufacturer_id	Product's Manufacturer ID	varchar(20)	xxxxxx				
	product_name	Product name	varchar(30)	Xxxxxxx				
	description	Product Description	varchar(100)	Xxxxxxx				
	product_price	Product price	decimal(8,2)	xxxx.xx		1 to 999999.99		
	number_of_product	Number of Product	int	xxxx		1 to 9999		
branch	branch_id	Branch ID	varchar(20)	xxxxxxx			PK	
	location	Branch Location	nvarchar(100)	Xxxxxxx				
	opening_hours	Branch's opening hour	datetime	xxxx-xx-xx xx:xx:xx				
	email	Branch's email	varchar(30)	Xxx@xx.com				
	phone_number	Branch's phone number	varchar(12)	0xx-xxx-xxxx				
review	review_id	Review ID	varchar(20)	xxxxxxx			PK	
	date	Review date	datetime	xxxx-xx-xx				
	rate	Review rate	decimal(2,1)	x				
	comment	Comment inside Review	nvarchar(200)	Xxxxxxxxxx				
	picture	Picture inside Review	varchar(200)	xxxxxxx.jpg	Y	1		
	customer_id	Customer ID	int	xxxxxxx			FK	Customer [Customer ID]
	product_id	Product's ID	varchar(20)	xxxxxx			FK	Product [Product ID]
order	order_id	Order ID	varchar(20)	xxxxxxx			PK	
	date	Ordered Date	date	xxxx-xx-xx				
	order_status	Order Status	varchar(200)	xxxxxxx				
	expected_delivery	Order Expected Delivery	date	xxxx-xx-xx				

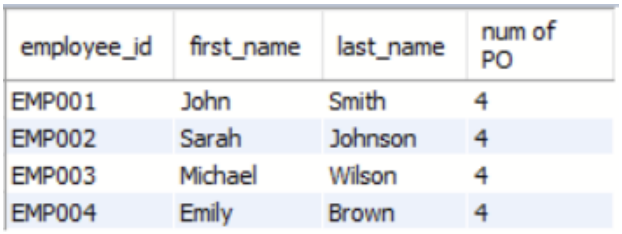
	customer_id	Customer's ID	varchar(20)	xxxxxxx			FK	Customer [Customer ID]
	branch_id	Branch's ID	varchar(20)	xxxxxxx			FK	Branch [ranch ID]
customer	customer_id	Customer's ID	varchar(20)	xxxxxxx			PK	
	first_name	Customer's First Name	nvarchar(30)	Xxxxxxx				
	last_name	Customer's Last Name	nvarchar(30)	Xxxxxxx				
	address	Customer's Address	nvarchar(100)	Xxxxxxx				
	phone_number	Customer's Phone Number	varchar(12)	0xx-xxx-xxxx				
	email	Customer's Email	varchar(30)	Xxx@xx.com				
	record_creation_date	Customer's Account Creation Date	date	xxxx-xx-xx				
individual	customer_id	Customer's ID	varchar(20)	xxxxxxx			PK, FK	Customer [Customer ID]
	session_id	Customer's session ID	varchar(20)	xxxxxxx				
	birth_date	Member's Birth Date	date	xxxx-xx-xx				
company	customer_id	Customer's ID	varchar(20)	xxxxxxx			PK, FK	Customer [Customer ID]
	company_id	Company ID	varchar(20)	xxxxxxx				
	company_name	Name of company	nvarchar(30)	xxxxxxx				

Query 21	products starting with 'S' (Upper case) and price > 20.
Owner	6788202 Chavanon Opaspakkthon
Objective	To Search for products starting with 'S' (Upper case) and price > 20.
Query	SELECT product_name, product_price FROM product WHERE product_name like 'S%' AND product_price > 20 ORDER BY product_price;
Result	

product_name	product_price
Soccer Ball	22.99
Super Hero Action Figure	24.99
Science Kit	34.99

Query 22	customer's first name, last name, and their account_age.
Owner	6788202 Chavanon Opaspakkthon
Objective	To show the customer's first name, last name, and their account_age.
Query	SELECT concat(first_name, ' ', last_name) AS customer_name, TIMESTAMPDIFF(YEAR, record_creation_date, CURDATE()) AS account_age FROM customer;
Result	

customer_name	account_age
James Wilson	1
Mary Thompson	1
Robert Garcia	1
Patricia Martinez	1
Michael Robinson	1
Jennifer Clark	1
William Rodriguez	1
Linda Lewis	1
David Lee	1
Barbara Walker	1
Richard Hall	1
Susan Allen	1

Query 23	Count POs by employees who have created more than 3 POs.
Owner	6788202 Chavanon Opaspakkthon
Objective	To count POs by employees who have created more than 3 POs.
Query	SELECT e.employee_id, e.first_name, e.last_name, count(p.po_id) AS "num of PO" FROM purchase_order p INNER JOIN employee e ON e.employee_id = p.employee_id GROUP BY employee_id HAVING count(po_id) > 3;
Result	

employee_id	first_name	last_name	num of PO
EMP001	John	Smith	4
EMP002	Sarah	Johnson	4
EMP003	Michael	Wilson	4
EMP004	Emily	Brown	4

Query 24	customer's first name, last name, and their expected delivery date.
Owner	6788202 Chavanon Opaspakkthon
Objective	To show the customer's first name, last name, and their expected delivery date.
Query	SELECT concat(c.first_name, ' ', c.last_name) AS customer_name, o.expected_delivery FROM customer c INNER JOIN order o ON o.customer_id = c.customer_id;
Result	

customer_name	expected_delivery
James Wilson	2025-11-05
Mary Thompson	2025-11-03
Robert Garcia	2025-11-08
Patricia Martinez	2025-11-06
Michael Robinson	2025-11-05
Jennifer Clark	2025-11-10
William Rodriguez	2025-11-09
Linda Lewis	2025-11-12
David Lee	2025-11-07
Barbara Walker	2025-11-11
Richard Hall	2025-11-13
Susan Allen	2025-11-09
James Wilson	2025-11-12
Robert Garcia	2025-11-09
Michael Robinson	2025-11-15
William Rodriguez	2025-11-13
David Lee	2025-11-11
Richard Hall	2025-11-16
Mary Thompson	2025-11-15
Patricia Martinez	2025-11-18
Jennifer Clark	2025-11-13
Linda Lewis	2025-11-17
Barbara Walker	2025-11-19
Susan Allen	2025-11-15
James Wilson	2025-11-18
Robert Garcia	2025-11-15
Michael Robinson	2025-11-21
William Rodriguez	2025-11-19
David Lee	2025-11-17
Richard Hall	2025-11-23
Mary Thompson	2025-11-20
Patricia Martinez	2025-11-18
Jennifer Clark	2025-11-24
Linda Lewis	2025-11-22
Barbara Walker	2025-11-20
Susan Allen	2025-11-25

Query 25	customer_id, first name, last name, and their order status.
Owner	6788202 Chavanon Opaspakkthon
Objective	To display customer_id, first name, last name, and their order status.
Query	SELECT c.customer_id, concat(c.first_name, ' ', c.last_name) AS customer_name, o.order_status FROM customer c LEFT JOIN order o ON o.customer_id = c.customer_id;
Result	

customer_id	customer_name	order_status
1001	James Wilson	Shipped
1001	James Wilson	Shipped
1001	James Wilson	Shipped
1002	Mary Thompson	Delivered
1002	Mary Thompson	Shipped
1002	Mary Thompson	Shipped
1003	Robert Garcia	Pending
1003	Robert Garcia	Delivered
1003	Robert Garcia	Delivered
1004	Patricia Martinez	Shipped
1004	Patricia Martinez	Pending
1004	Patricia Martinez	Delivered
1005	Michael Robinson	Delivered
1005	Michael Robinson	Pending
1005	Michael Robinson	Pending
1006	Jennifer Clark	Cancelled
1006	Jennifer Clark	Delivered
1006	Jennifer Clark	Pending
1007	William Rodriguez	Shipped
1007	William Rodriguez	Shipped
1008	Linda Lewis	Pending
1008	Linda Lewis	Shipped
1008	Linda Lewis	Shipped
1009	David Lee	Delivered
1009	David Lee	Delivered
1009	David Lee	Delivered
1010	Barbara Walker	Shipped
1010	Barbara Walker	Pending
1010	Barbara Walker	Delivered
1011	Richard Hall	Pending
1011	Richard Hall	Cancelled
1011	Richard Hall	Pending
1012	Susan Allen	Delivered
1012	Susan Allen	Delivered
1012	Susan Allen	Cancelled